

Thickness Speed Progression Index: Machine Learning Approach for Keratoconus Detection



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- **PURPOSE:** To develop and validate a pachymetry-based machine learning (ML) index for differentiating keratoconus, keratoconus suspect, and normal corneas.
- **DESIGN:** Development and validation of an ML diagnostic algorithm.
- **METHODS:** This retrospective study included 349 eyes of 349 patients with normal, frank keratoconus (KC), and KC suspect (KCS) corneas. KCS corneas included topographically/tomographically normal (TNF) and borderline fellow eyes (TBF) of patients with asymmetric KC. Six parameters were derived from the corneal thickness progression map on the Galilei Dual Scheimpflug-Placido system and fed into a machine-learning algorithm to create the Thickness Speed Progression Index. The model was trained with 5-fold cross-validation using a random search over 7 different ML algorithms, and the best model and hyperparameters were selected.
- **RESULTS:** A total of 133 normal eyes, 141 KC eyes, and 75 KCS eyes, subdivided into 34 TNF and 41 TBF eyes, were included. In experiment 1 (normal and KC), the best model (Random Forest) achieved an accuracy of 100% and area under the receiver operating characteristic (AUROC) of 1.00 for both normal and KC groups. In experiment 2 (normal, KCS, and KC), the model achieved an overall accuracy of 91%, and AUROC curves of 0.93, 0.83, and 0.99 in detecting normal, KCS, and KC corneas respectively. In experiment 3 (normal, TNF, TBF, and KC), the model achieved an accuracy of 87% with AUROC curves of 0.91, 0.60, 0.77, and 0.94 for normal, TNF, TBF, and KC corneas, respectively.

 Supplemental Material available at [AJO.com](https://ajocomm.org).
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- **CONCLUSIONS:** Using data solely based on pachymetry, ML algorithms such as the Thickness Speed Progression Index are able to discriminate normal corneas from KC and KCSs corneas with reasonable accuracy. (Am J Ophthalmol 2025;271: 188–201. © 2024 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>))

INTRODUCTION

THE EARLY DETECTION OF KERATOCONUS (KC) REMAINS critically important, particularly given the popularity of refractive surgery and concerns for postrefractive surgery ectasia development.^{1,2} The latest endeavors have focused on the use of corneal biomechanical properties for the detection of ectasia susceptibility³⁻⁷ However, current clinically available methods for measuring the biomechanical strength of the cornea,^{8,9} despite being somewhat successful, have their limitations, most importantly their inability to detect asymmetry, the hallmark of KC, in the corneal profile. Conventional methods for detecting KC corneas remain a mainstay of clinical practice and have proven effective, particularly when information about different aspects of the corneal profile, often provided by more than one imaging modality, is combined.¹⁰ However, pachymetry is a cornerstone of KC diagnosis, since corneal thinning can be the earliest sign of KC,¹¹ and unlike curvature and elevation, is less likely to be dependent on fixation.¹²

Several findings within a corneal pachymetric profile raise concern for a keratoconic cornea, or a cornea with an underlying risk for ectasia. A thin cornea, especially if less than 480 μm thick, and a displaced thinnest point, are concerns for potential abnormality.^{13,14} However, conclusions cannot be drawn solely based on such findings, as both markers may be normal variants and can therefore be nonspecific. The rate of pachymetric progression from the thinnest point to the periphery was introduced by Ambrosio et al.¹⁵ and has been found to differ between normal corneas and keratoconic corneas, even those with mild forms of the disease.¹⁶⁻²⁰ Based on these findings,

several parameters have been developed and have proven somewhat useful in clinical evaluations. However, to the best of our knowledge, all currently available parameters assess pachymetric progression using annular/concentric averages of thickness, or meridional averages.^{15,17,18} Concentric averages can attenuate findings and fail to reveal the asymmetric properties of an ectatic cornea. Meridional averages may also attenuate, and therefore fail to detect, early focal changes that may not involve the entire radius of the cornea.

To overcome the limitation of data dilution brought about by using such meridional and annular averages, we propose a method that would evaluate all points on the pachymetric map, generating parameters located at different angles and radii from the thinnest point. The generated parameters are fed into a machine-learning algorithm to create the Thickness Speed Progression Index (TSPI), an index based solely on pachymetry for the detection of KC- and KC-susceptible corneas.

METHODS

This retrospective observational study was conducted in the Cornea and Refractive Surgery Division of the Department of Ophthalmology at the American University of Beirut Medical Center in Beirut, Lebanon. Approval was obtained from the Institutional Review Board (IRB ID: BIO-2019-0399), and the study adhered to the tenets of the Declaration of Helsinki.

- **PATIENT SELECTION:** This study included patients with normal control corneas, patients with frank KC in both eyes and patients with KC suspect (KCS) corneas. All patients had previously undergone imaging with the Galilei dual Scheimpflug-Placido topography/tomography system (Ziemer Group AG) as part of their routine screening for refractive surgery, or follow-up for KC. Exclusion criteria were inadequate Galilei image acquisitions, contact lens wear less than 2 weeks prior to image acquisition, the presence of corneal scarring, a history of corneal or ocular surgery, such as laser in situ keratomileusis, corneal cross-linking or intrastromal ring insertion, as well as other ocular pathologies such as retinal disease.

KC was defined using both objective and subjective criteria, as previously described by the authors.^{21,22} The objective criterion used was based on the Cone Location and Magnitude Index (CLMI.X), an upgraded version of the CLMI that was extended to include data not only from the anterior surface but also from the pachymetry and posterior surface. It has a sensitivity of 99.4% and a specificity of 99.6% specificity in identifying KC.²³ All KC corneas had a CLMI.X of 100%. For subjective assessment, criteria considered findings and indices describing anterior corneal curvature asymmetry and posterior surface abnormalities,

and the final decision was made by a refractive surgeon based on the combination of these findings. Major criteria for the former included a skewed radial axis exceeding 20°, an inferior-superior value above 2.38 diopters (D), and a KC Prediction Index (KPI) above 30%. Minor criteria included inferior steepening of at least 1.00 D, an asymmetrical bowtie pattern without a skewed radial axis, an inferior-superior value of over 1.40 D but below 2.30 D, and a KPI between 18.5% and 30%.²⁴ Posterior surface abnormalities were identified by a highest posterior elevation point exceeding 18 µm in the 3-mm central zone of best-fit sphere or over 22.5 µm in the 5-mm central zone.²⁵ The subjective diagnosis of KC relied on at least one major criterion or two minor criteria together with a posterior surface abnormality.²²

Normal control corneas were those of patients with no clinical, topographic, or tomographic evidence of KC in either eye, in addition to no family history of KC. All normal control patients underwent bilateral laser in situ keratomileusis and had no evidence of ectasia at least 3 years following the procedure. The selection of which eye (left or right) to use for normal and KC patients was made randomly.

KCS corneas were the fellow eyes of highly asymmetric KC patients. These corneas failed to meet the previously defined criteria of KC, but their contralateral eye was diagnosed with the disease. These eyes have commonly been described as forme fruste KC, based only on normal topographic criteria.²⁶ However, as previously described by our group,²² we have extended this definition to include both topographic and tomographic criteria, and have subdivided KCS eyes as follows: tomographically and topographically normal fellow eyes of KC patients (TNF) and tomographically and topographically borderline fellow eyes (TBF). TNF eyes failed to meet the diagnosis of KC, both objectively (CLMI.X = 0%) and on subjective assessment, while they met the following objective criteria: KPI < 10%^{24,27,28} and Asphericity Asymmetry Index < 21.5 mm.^{29,30} TBF eyes had at least one of the following: 10% < KPI < 18.5%,^{24,27,28} and 21.5 < posterior Asphericity Asymmetry Index < 34.5 mm.^{29,30} Both TNF and TBF eyes had a CLMI.X < 25%.²³

- **DATA PREPARATION:** The corneal thickness progression (CTP) data used in this study was derived from the Galilei dual Scheimpflug-Placido system. It is based on the radial speed of thickness progression, expressed in mm/mm, of every point on the cornea relative to the thinnest point (Eq. 1). As opposed to the average change, the CTP is the local derivative of the pachymetry map in the radial direction.

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \quad (1)$$

Where x is the radius (in mm), and y is the corneal thickness (in mm) at any given point.

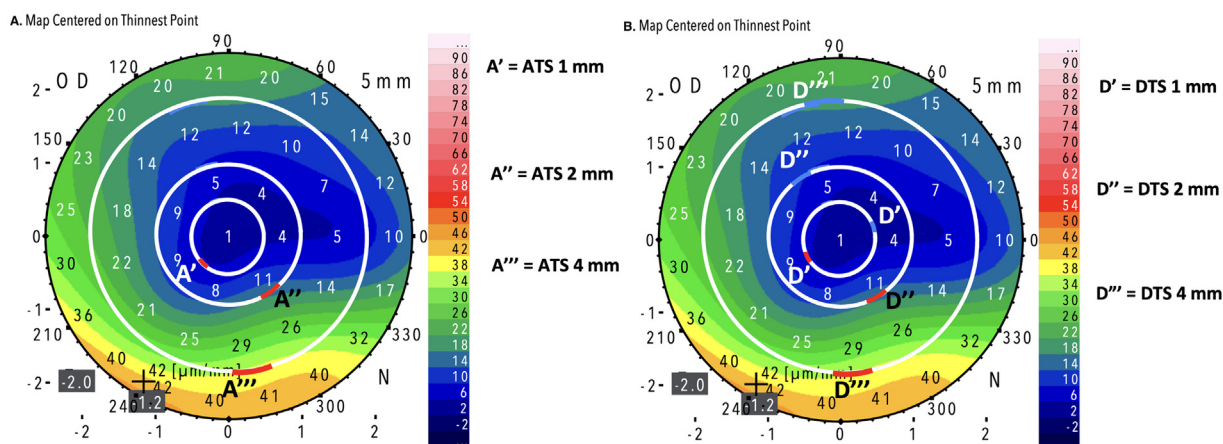


FIGURE 1. Corneal thickness progression map with sector division showing the 15-degree arcs at 1.0-, 2.0-, and 4.0-mm diameters. (A) The red arcs represent the greatest absolute thickness speed (ATS). (B) The greatest difference between diametrically opposite red and blue arcs is the difference thickness speed (DTS). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

TABLE 1. Patient Demographics and Ocular Parameters

Parameter	Mean $\pm$ Standard Deviation (Range)				P Value
	NL	TNF	TBF	KC	
Number of eyes n (%)	133 (38.1%)	34 (9.7%)	41 (11.7%)	141 (40.5%)	—
Age	37.22 $\pm$ 12.30 (19; 70)	35.29 $\pm$ 12.34 (21; 78)	35.20 $\pm$ 12.97 (22; 74)	35.70 $\pm$ 12.65 (11; 73)	.478
Sex (% males)	47%	42%	61%	59%	.122
MR sphere (D)	−1.90 $\pm$ 2.86 (−11.00; 6.75)	−1.97 $\pm$ 2.26 (−8.75; 2.50)	−2.03 $\pm$ 2.24 (−8.00; 2.50)	−1.73 $\pm$ 2.58 (−11.50; 4.25)	.644
MR Cyl (D)	−1.47 $\pm$ 1.73 (−10.00; 3.00)	−1.61 $\pm$ 1.47 (−5.75; 0.00)	−1.33 $\pm$ 1.46 (−8.00; 0.00)	−1.35 $\pm$ 1.40 (−6.00; 3.00)	.762
MRSE (D)	−2.31 $\pm$ 2.81 (−11.00; 4.63)	−2.55 $\pm$ 2.38 (−10.13; 2.50)	−2.56 $\pm$ 2.49 (−12.00; 2.00)	−2.04 $\pm$ 2.58 (−11.88; 3.88)	.216
KMean (D)	43.78 $\pm$ 1.90 (39.01; 48.56)	44.67 $\pm$ 1.54 (42.13; 47.92)	43.76 $\pm$ 1.81 (39.97; 47.20)	44.62 $\pm$ 2.04 (40.52; 52.67)	.003
CCT (microns)	550.24 $\pm$ 30.84 (498.00; 645.00)	517.09 $\pm$ 36.79 (443.00; 577.00)	512.63 $\pm$ 32.285 (425.00; 566.00)	473.61 $\pm$ 45.26 (323.00; 589.00)	<.001
KPI (%)	1.71 $\pm$ 3.04 (0; 9.2)	1.64 $\pm$ 2.47 (0; 7.90)	5.23 $\pm$ 5.72 (0; 18.10)	57.50 $\pm$ 30.75 (14.43; 100)	.007
AAI (%)	16.47 $\pm$ 3.87 (7.50; 21.00)	14.45 $\pm$ 4.03 (5.00; 21.00)	28.28 $\pm$ 3.34 (21.00; 33.00)	86.75 $\pm$ 44.55 (24.10; 186.50)	<.001

Significance is defined as p-value less than or equal to 0.05.

AAI = Asphericity Asymmetry Index; CCT = central corneal thickness; KPI = Keratoconus Prediction Index; MR = manifest refraction; MRSE = manifest refractive spherical equivalent.

Using the data export feature on Galilei (available through the CSV export license), the CTP CSV files of all included patients were exported. The exported data contained 18,001 points, giving their location in polar coordinates in the form of radius (in mm) and angle (in degrees), as well as their speed of thickness progression value (mm/mm). The points are arranged in 50 concentric circles centered on the corneal vertex, with increasing radii of 0.1 to 5.0 mm in 0.1 mm increments, and 360 points of thickness progression speed distributed equally on each circle.

The map was transposed such that all points were centered on the thinnest point (coordinates [0, 0°]). Only points within the central 6.0 mm diameter circle of the cornea, centered on the thinnest point, were included in the analysis to avoid peripheral artifacts such as those caused by patients' eyelids or tear film pooling.

• **ABSOLUTE THICKNESS SPEED (ATS) AND DIFFERENCE THICKNESS SPEED (DTS) PARAMETERS:** Averages of the speed of thickness progression were calculated for every 15-

TABLE 2. Results of the Different Experiments

Experiment	Best Model	Class	Accuracy (%)	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)	F1-score	AUROC
Exp (1)	All models except LR	NL	100	100	100	100	100	1	1
		KC		100	100	100	100	1	
Exp (2)	Random forest	NL	91	96	89	84	97	0.90	0.93
		KCS		68	97	86	92	0.76	0.83
		KC		98	100	100	99	0.99	0.99
Exp (3)	Random forest	NL	87	93	88	83	96	0.88	0.91
		TNF		41	97	56	94	0.47	0.60
		TBF		83	97	79	98	0.81	0.77
		KC		94	100	100	96	0.96	0.94

Experiment 1 compares normal (NL) and keratoconus (KC) eyes. Experiment 2 compares NL, keratoconus susceptible (KCS), and KC eyes. Experiment 3 compares NL, tomographically/topographically normal (TNF), topographically/topographically borderline (TBF), and KC eyes.

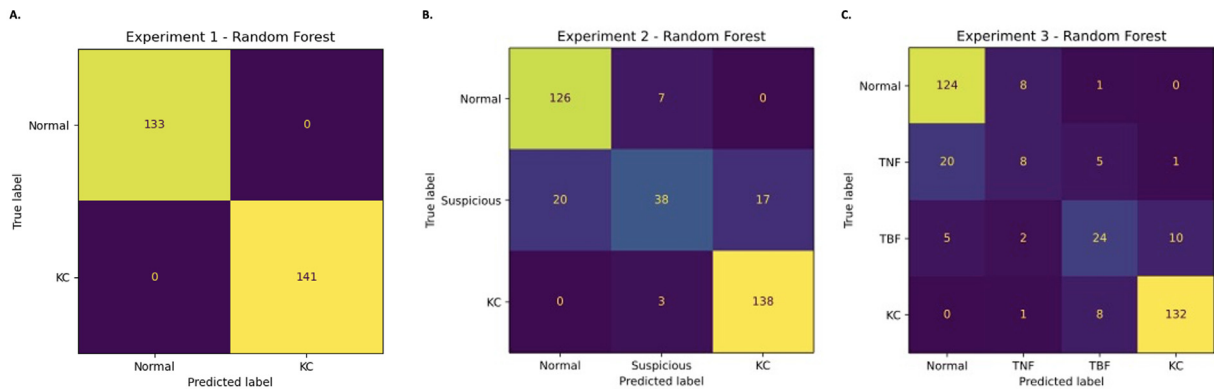


FIGURE 2. Confusion matrices of TSPI. (A) Confusion matrix of experiment 1 (NL and KC eyes). (B) Confusion matrix of experiment 2 (NL, KCS, and KC eyes). (C) Confusion matrix of experiment 3 (NL, TNF, TBF, and KC eyes). There were no false positives or false negatives between NL and KC groups.

degree arc with radii of 0.5, 1.0, 1.5, 2.0, 2.5, and 3.0 mm (diameters of 1.0, 2.0, 3.0, 4.0, 5.0 and 6.0 mm, respectively). The arcs with the highest averages were determined for each of the radii, and the greatest differences in averaged speed of thickness progression between diametrically opposite 15-degree arcs were determined at each radius (Figure 1). These parameters were named absolute thickness speed (ATS) and DTS, respectively. The area under the receiver operating characteristic (AUROC) curve was calculated to differentiate normal from KC eyes. The AUROC curves were greatest for ATS and DTS at diameters of 1.0, 2.0, and 4.0 mm. Hence, 6 parameters, ATS1.0, ATS2.0, ATS4.0, DTS1.0, DTS2.0, and DTS4.0 were chosen to train the machine learning (ML) models to distinguish between normal and KC corneas. Additionally, for each of these parameters, AUROC curve analysis was performed for normal versus KC eyes, normal versus TBF eyes, and normal versus TNF eyes.

• **ML MODELS:** For each eye, the 6 variables (ATS1.0, ATS2.0, ATS4.0, DTS1.0, DTS2.0, and DTS4.0) were collected and used to train the models.

First, standard scaling was applied to all variables by subtracting the mean and scaling to unit variance to allow

faster convergence of the models during the learning phase. An isolation forest, a type of unsupervised ML for anomaly detection, was used for KCS patients to identify outliers and make sure they are not included in the training set to avoid building a skewed model. Next, the data were divided into a training set (80%), used by the models to learn relevant relations between the input and the output, and a testing set (20%), used to evaluate the performance of the models. To account for the imbalance in proportions of the outcome (more normal and KC corneas compared to KCS corneas), which might lead to biased models, the synthetic minority oversampling technique³¹ was applied to the KCS class in the training set.

To build the models, six state-of-the-art ML algorithms were implemented and tested, in addition to the conventional statistical modeling technique (Logistic Regression). The models used were Support Vector Machine, K-Nearest Neighbor, Decision Tree, Random Forest (RF), XGBoost, and CatBoost. For each algorithm, a random search was performed to find the best set of hyperparameters (tunable parameters that are inherent to the model). This technique evaluates randomly the hyperparameter space and finds the optimal combination in a computationally efficient way. To

ensure that the obtained models were generalizable and not sensitive to the data split, 5-fold cross-validation was performed at each step for every algorithm.

The performance of the models was evaluated using accuracy, sensitivity, specificity, positive predictive value, negative predictive value, F1-score, and AUROC curve.

TNF eyes that were predicted as TBF or KC were considered correct, as were TBF eyes predicted as KC. This was done to favorize the detection of borderline cases. The best model was selected based on overall higher average accuracy and AUROC curve. All steps were implemented using the Imbalanced learn pipeline and the Scikit-learn

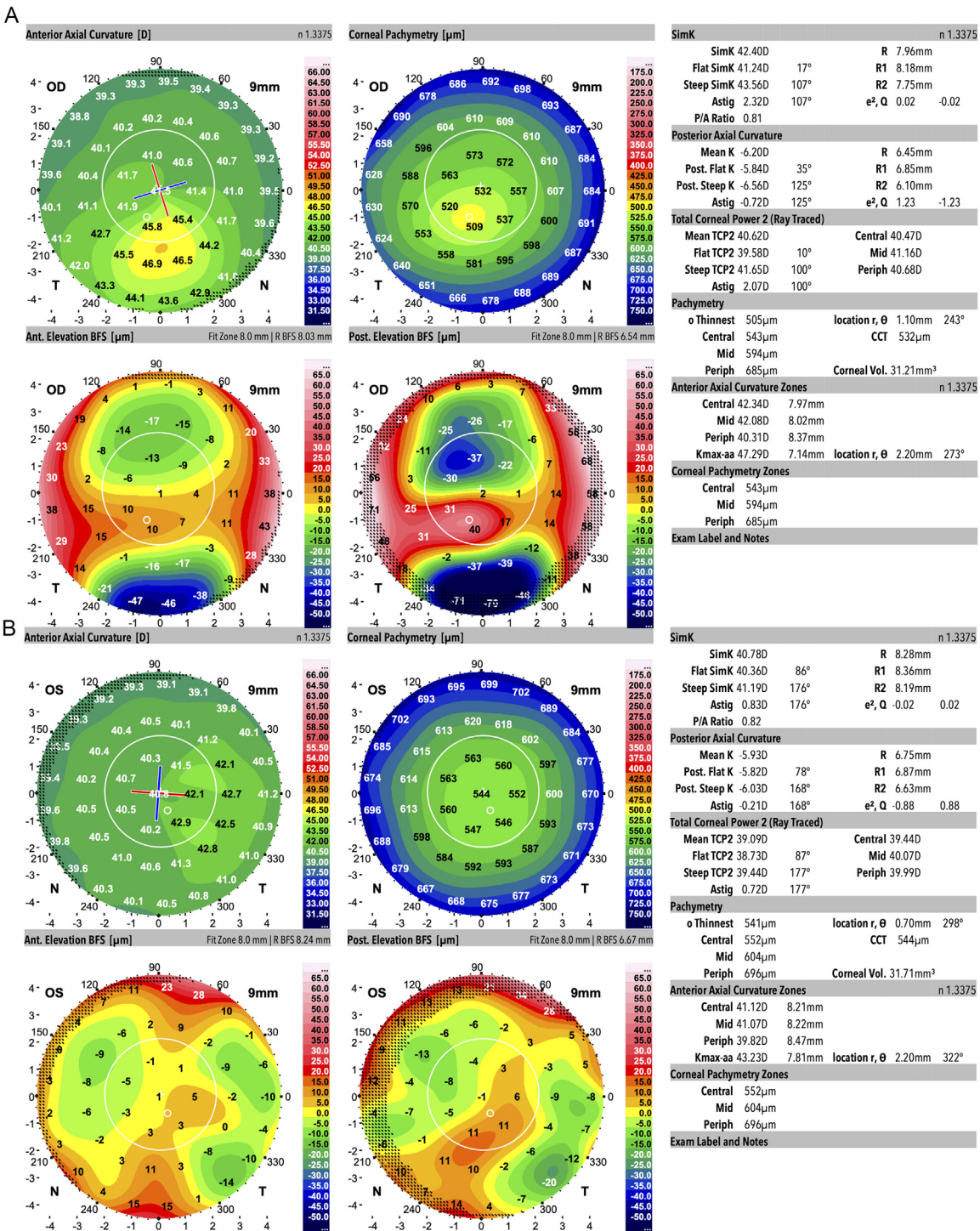


FIGURE 3. Dual Scheimpflug/Placido imaging (Galilei) of a patient with part (A) diagnosed keratoconus in the right eye and part (B) the fellow left eye with no definitive abnormality that would diagnose keratoconus, and part (C) a normal CLMI.X (OD) at 0%.

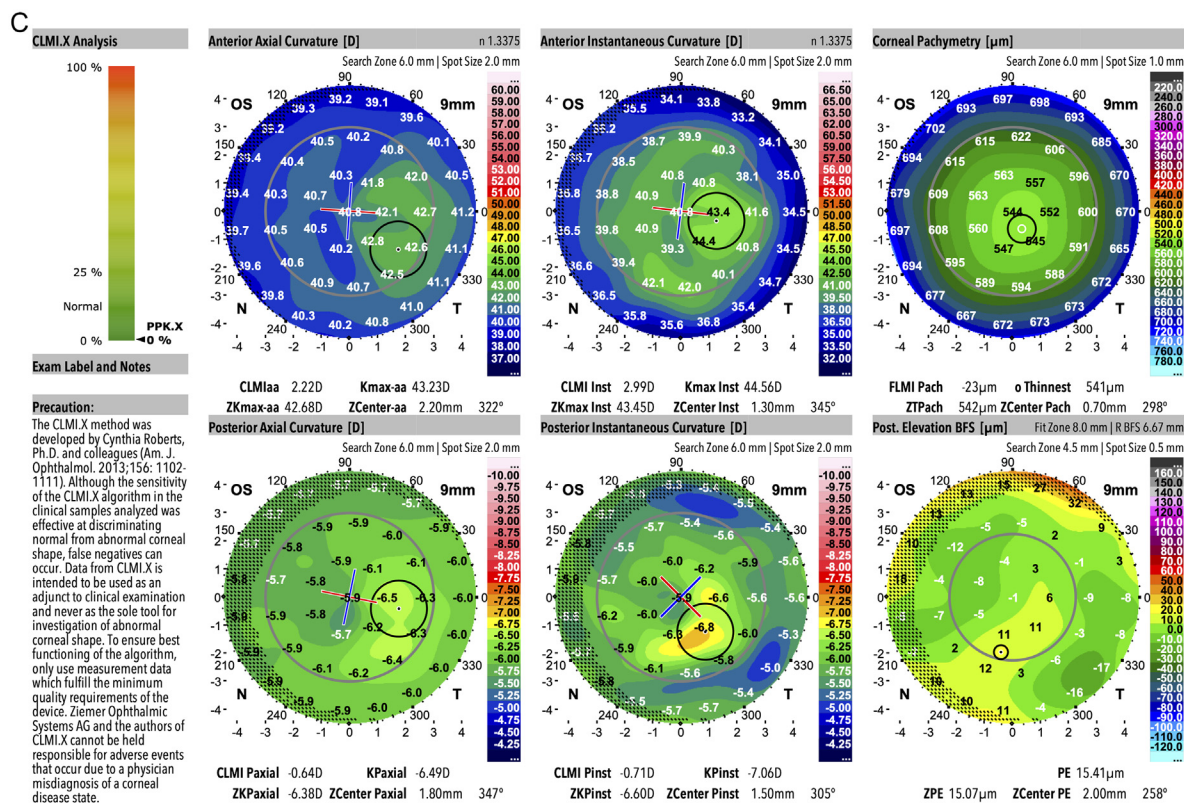


FIGURE 3. Continued

library on Python 3.7. The final model was named the TSPI.

• **EXPERIMENTS:** Using the previous workflow, 3 different experiments were conducted by training the models using:

- (1) Normal and KC corneas in Experiment 1
- (2) Normal, KCS, and KC corneas in Experiment 2
- (3) Normal, TNF, TBF, and KC corneas in Experiment 3.

• **STATISTICAL ANALYSIS:** For categorical variables, frequencies and proportions were reported and statistical significance was tested using Chi-square. For continuous variables, means and standard deviations were reported. Shapiro–Wilk test was used to test for normality, and statistical significance for different groups was tested using analysis of variance for normally distributed variables, and Kruskal–Wallis for non-normally distributed variables. *P* values of less than .05 were considered significant. All statistical analyses were performed using IBM SPSS Statistics software version 26.

RESULTS

This study included 349 eyes from 349 patients: 133 eyes with normal control corneas, 141 eyes with frank KC, and

75 eyes with KCS corneas, of which 34 eyes were TNF eyes of patients with diagnosed KC in the other eye, and 41 eyes were TBF eyes. Of the 141 KC eyes, 136 (96.5%) were classified as grade 1 or grade 2 according to the Amsler–Krumeich classification. Patient demographics and other ocular parameters are summarized in Table 1.

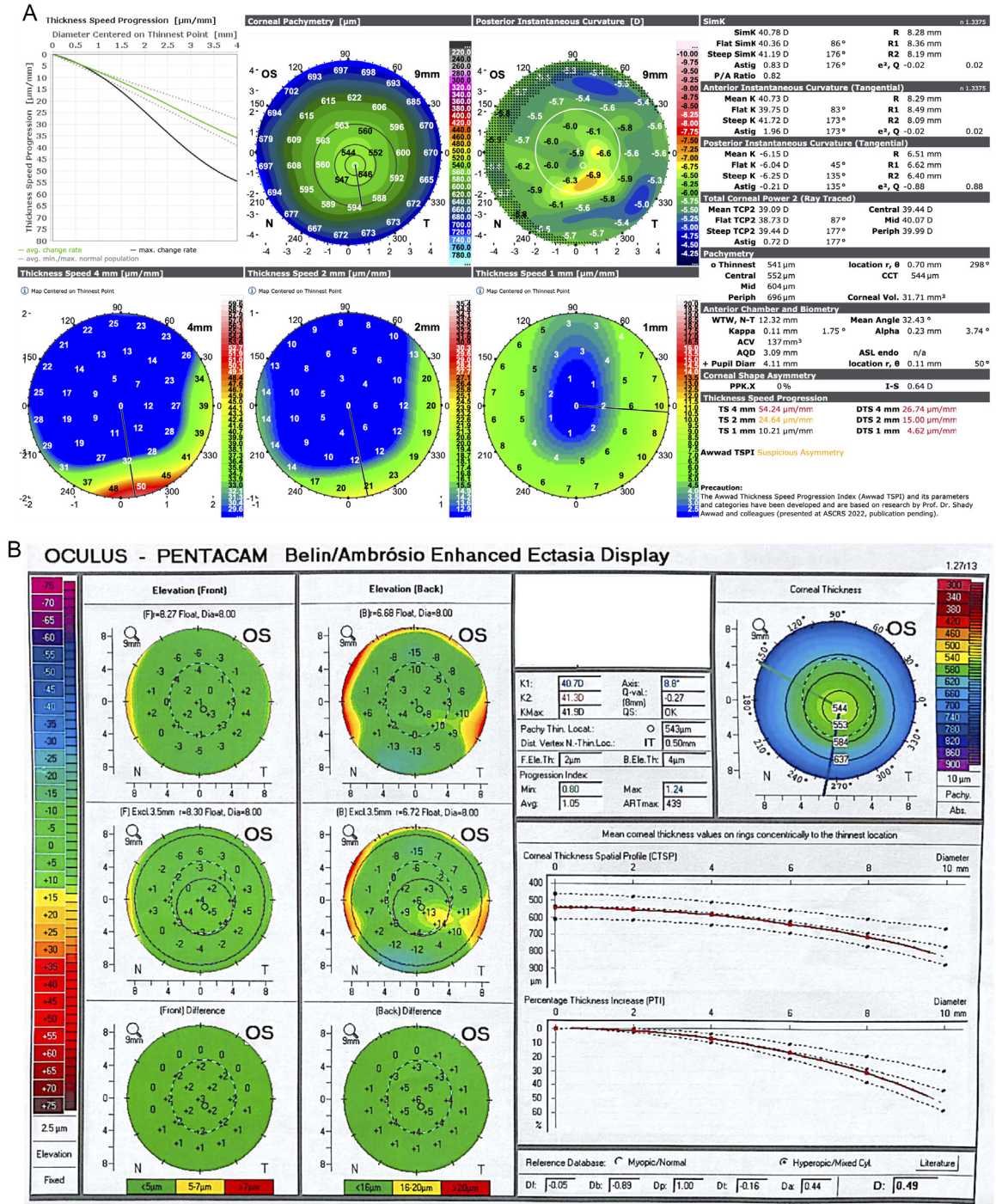
The Supplemental Material displays the ROC curves for each of the 6 features selected (ATS1.0, ATS2.0, ATS4.0, DTS1.0, DTS2.0, and DTS4.0), when used for differentiating normal from KC corneas. As shown (Supplemental Material—Figure A), the AUROC curves were 0.89 for DTS1.0, 0.93 for DTS2.0, 0.94 for DTS4.0, and 1.00 for ATS1.0, ATS2.0, and ATS4.0. ROC curves for the differentiation of normal from TBF eyes (AUROC curves range: 0.76-0.94) and normal from TNF eyes (AUROC curves range: 0.55-0.68) are plotted in (Supplemental Material—Figures B and C), respectively.

Table 2 summarizes the results obtained by the best models in each experiment. For Experiment 1 (Normal vs KC), all the ML models were able to differentiate KC from normal corneas with 100% accuracy and an AUROC curve of 1 (Figure 2, A), while the Logistic Regression model had an accuracy of 99% and an AUROC curve of 0.99. For experiments 2 and 3, the best-performing model was the RF model; a widely used algorithm trained by constructing a group of decision trees using different subsets of the data and averaging their predictions to minimize overfitting and improve generalizability. For Experiment 2 (Normal vs KCS vs

KC), the RF model (Figure 2, B) was able to detect Normal, KCS, and KC corneas with 91% accuracy and AUROC curves of 0.93, 0.83, and 0.99, respectively (Table 2). For Experiment 3 (Normal vs TNF vs TBF vs KC), the model (RF, Figure 2, C) was able to detect Normal, TNF, TBF, and

KC corneas with an accuracy of 87% and AUROC curves of 0.91, 0.60, 0.77, and 0.94, respectively (Table 2).

To showcase the importance of the TSPI for the detection of KCS eyes when compared to other available parameters, Figures 3 to 6, present data from two patients, each



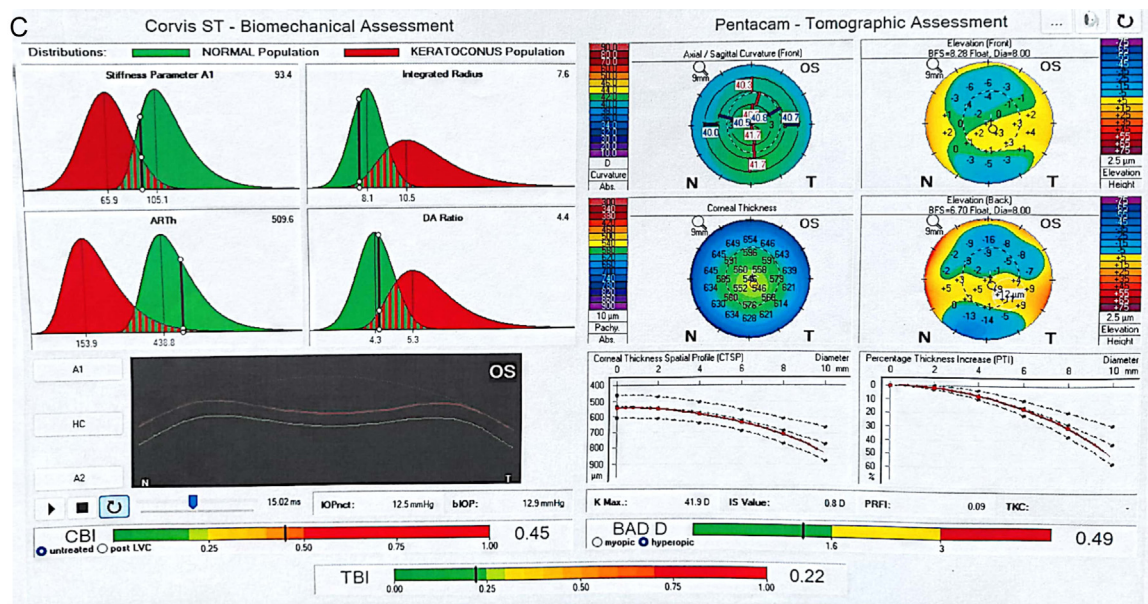


FIGURE 4. Continued

with the diagnosis of KC in one eye, with some asymmetric changes in the fellow eye that fail to meet the diagnosis of KC. Figures 3 and 4 display the Galilei, Pentacam, and Corvis data of a patient with diagnosed KC in the left eye and a fellow right eye that fails to meet criteria for the diagnosis of KC. The TSPI labels the right fellow eye as having “suspicious asymmetry,” thus indicating that it is a KCS eye. However, the pachymetric progression indices of the Pentacam as displayed on the Belin/Ambrósio Enhanced Ectasia Display are all normal. Interestingly, the SP-A1 and TBI also fail to characterize this cornea as being at risk of ectasia; however, a CBI of 0.49 appropriately designates it as a cornea at moderate risk for ectasia.

Figure 5 displays the Galilei data of the left fellow eye (Figure 5, A) of a patient with diagnosed KC in the right eye. This eye fails to meet the criteria for the diagnosis of KC, has a normal CLMI.X (Figure 5, B) but was labeled by the TSPI as having “suspicious asymmetry,” and thus a KCS (Figure 5, C). Galilei images for the same patient, captured 3 years later, are displayed in Figure 6, showing a slightly increased steepening, thinning, and elevation (Figure 6, A), an abnormal CLMI.X. X of 47% (Figure 6, B), and a TSPI predicting “pathologic asymmetry” (Figure 6, C).

DISCUSSION

The TSPI, an ML model based only on CTP, without any anterior or posterior curvature or elevation data, was able to differentiate normal, KC susceptible, and frank KC corneas with reasonable accuracy.

Our analysis of TSPI revealed no false positives or false negatives among normal and KC eyes. This demonstrates the index’s robustness in the detection of KC, including its mild forms that comprised the bulk of our dataset, even when used as the sole diagnostic marker. The index has a lower accuracy in detecting TBF eyes, which is a vital feature in screening for candidacy for laser vision correction, and this is consistent with the literature with other approaches or indices, which typically use multiple and variable parameters. The accuracy is also lower for the TNF group of eyes, a category that has historically been very difficult to discriminate.

When comparing to our previous work on the role of biomechanics in detecting TBF and TNF, the TSPI had similar performance to the Tomography Biomechanical Index 1.0 (TBI 1.0): 91% sensitivity and 95% specificity for TBF eyes, and 54% sensitivity and 96% specificity for TNF eyes.²² The Corvis Biomechanical Index (CBI), which mainly relies on biomechanical parameters, yielded 61% sensitivity and 96% specificity for TBF eyes and 29% and 97%, respectively for TNF eyes.²² Pentacam and Corvis data were not available for all members of the cohort, and hence, a direct head-to-head comparison was not possible.

In comparison to the performance of other pachymetry-based indices for the detection of frank KC, the TSPI (AUROC curve: 1.0) outperformed the ARTh,³²⁻³⁴ ARTmin,¹⁹ ARTavg,^{17,19,20} ARTmax,^{17,19,20,24,34} PPImin,²⁰ PPIavg,^{17,19,20} and PPImax,^{17,19,20,34} that had AUROC curves of less than 1.0 in the comparison of normal and KC eyes.

For the detection of KCSs however, the results in comparison to the performance of the TSPI are conflicting. The TSPI in our study (AUROC curve: 0.77) outper-

formed ARTh (AUROC curve: 0.61133 and 0.71834, ARTmin (AUROC curve: 0.5619), ARTavg (AUROC curve: 0.5719), PPIavg (AUROC curve: 0.40419), and PPI-max (AUROC curve: 0.6419) in some studies. In other

studies, ARTavg,²⁰ ARTmax,^{19,20,24,34} PPImin,²⁰ PPIavg,²⁰ and PPImax^{20,34} had greater AUROC curves than the TSPI for the identification of suspect eyes. However, in all the aforementioned studies, the criteria used for selecting

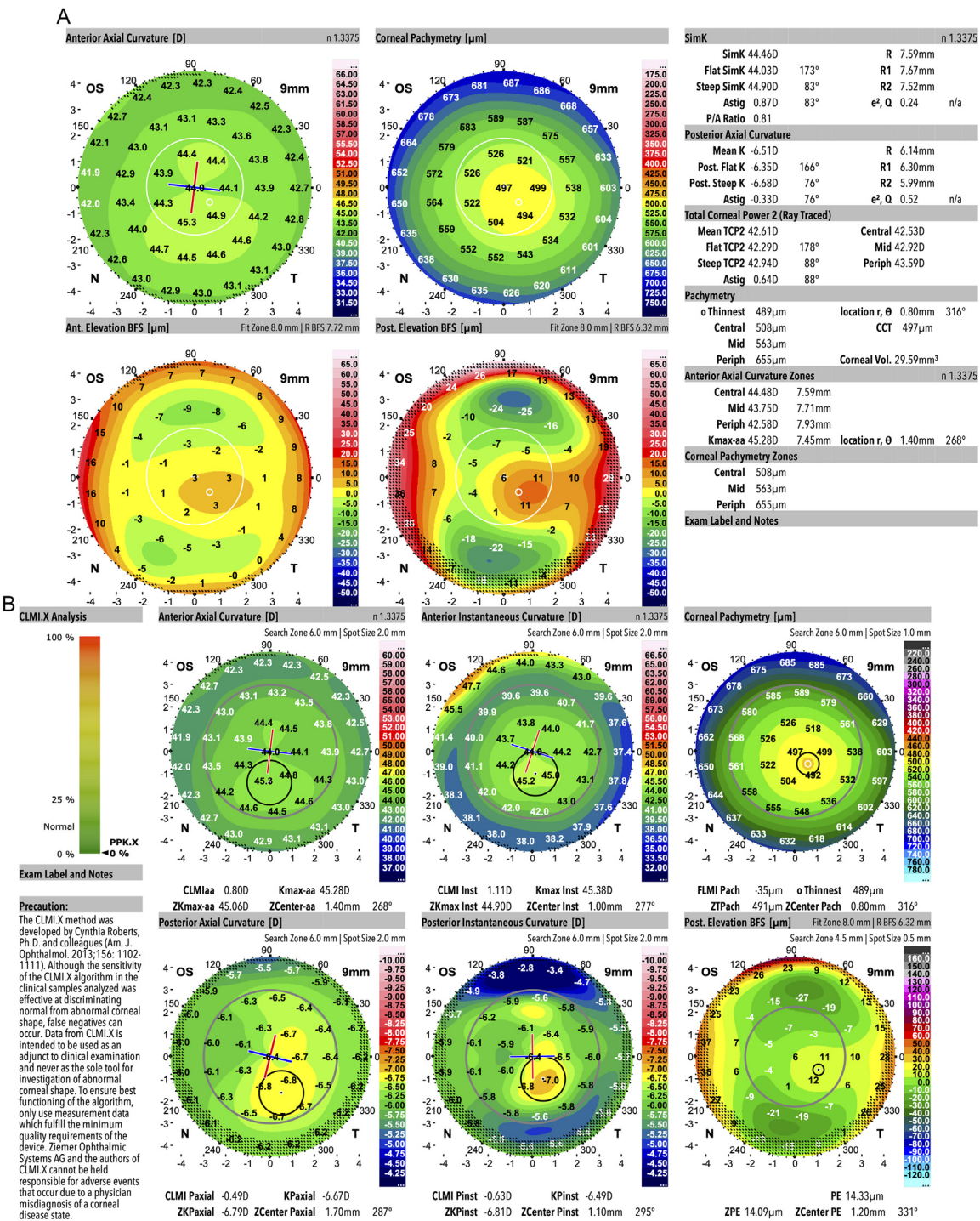


FIGURE 5. (A) Dual Scheimpflug/Placido imaging (Galilei) of the fellow eye (OS) of a patient with diagnosed keratoconus in the contralateral eye (OD, not shown). There is no definitive abnormality that would diagnose keratoconus in this eye. (B) CLMI-X analysis was normal at 0%, indicating a normal, nonectatic cornea. (C) The TSPI reveals a suspicious pachymetry rate change, thus correctly labeling this eye as keratoconus suspect.

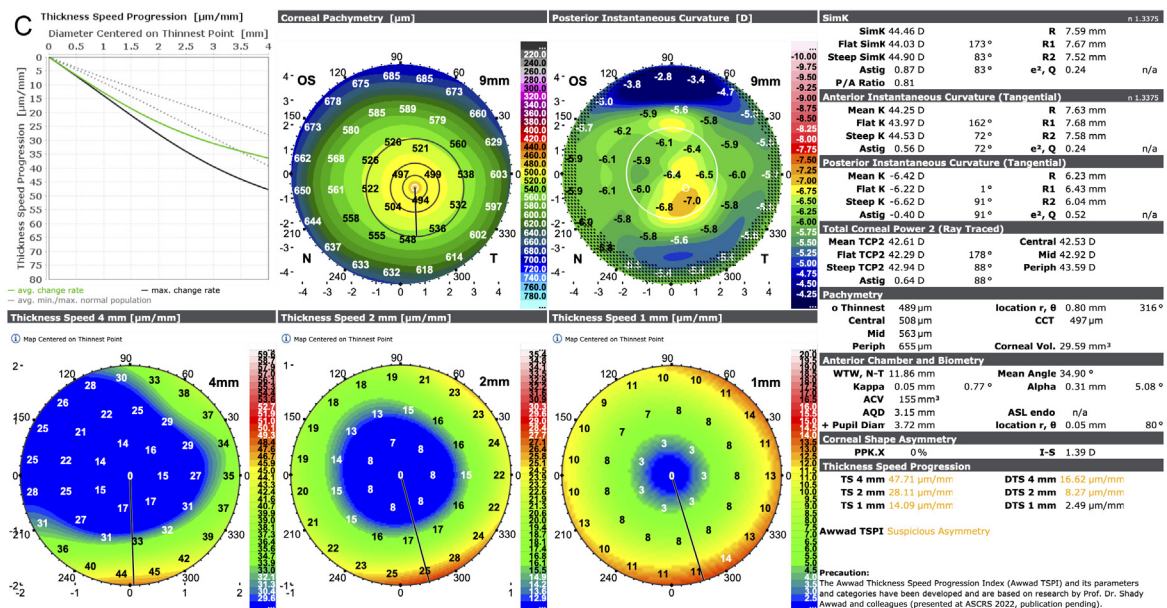


FIGURE 5. Continued

KCSs, subclinical KC, or forme fruste KC were considerably less stringent than our criteria, making their identification as abnormal less challenging. That said, the aim of this work is not to show superiority of the TSPI over other modalities or indices, but rather to showcase the potential of the TSPI in maximizing the analysis of the pachymetry map, and overcoming the limitation of the averaging-induced dilution.

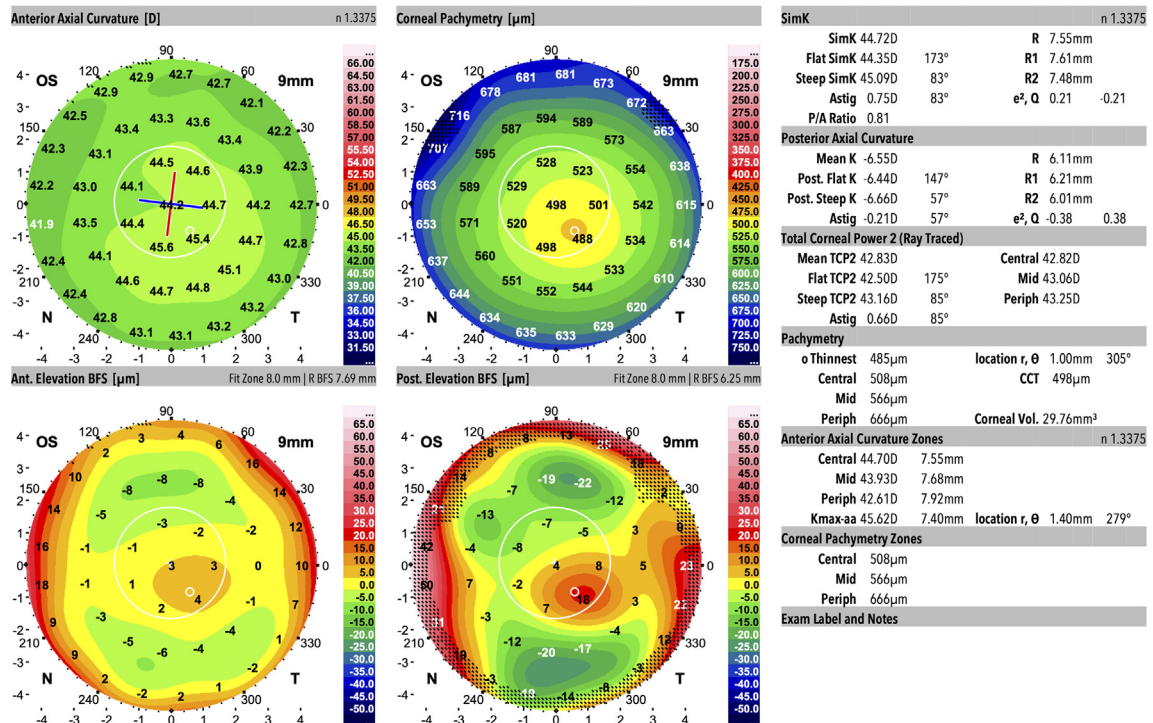
The TSPI was built using the local derivative of the pachymetry map in the radial direction centered on the thinnest point. The local derivative approach enables the detection of local pachymetric change and asymmetry, and centration on the thinnest point allows for quick visual interpretation of a derivative map with only positive values. The power of this approach is that it does not average same-distance data from the thinnest points over 360° to generate a progression map, as has been the trend.¹⁸ The latter approach, while being instrumental in highlighting the importance of thickness progression in KC detection, has the potential to attenuate early and mild focal changes in pachymetric progression, as these changes would be averaged with the rest of the data values at the same distance from the thinnest point, but at different angles. Locating the fastest and slowest hemi-meridian, as has been described before with the PPImax and PPImin,³⁵ is still not ideal, as it also suffers from possible attenuation of early and mild focal data along the given hemi-meridian. Our approach uses pachymetry data along all the meridians and over different radii, which minimizes the possibility of attenuating early focal changes in pachymetric progression, which maximizes their detection. These findings are corroborated by the data presented in Figures 3 to 6, which clearly show that

current tomography- and topography-based indices fail to accurately predict the risk of ectasia in the fellow eye, except for the TSPI and a single biomechanical index (CBI). This suggests that early biomechanical changes alongside subtle focal changes in pachymetric progression may occur before other more apparent morphological alterations take place. This feature is particularly vital in the detection of suspicious and subclinical KC. Utilizing all pachymetry data points across all meridians and radii would have been optimal, albeit necessitating an extensive dataset to train a supervised deep learning model. In light of the limited availability of TNF and TBF eyes in the clinical setting, we found that our approach of preselecting parameters with high AUROC curves and subsequently inputting them into various ML models was the most efficient methodology.

Moreover, pachymetric progression data centered on the thinnest point is potentially less likely to be dependent on fixation as opposed to curvature and elevation data. The latter has been shown to be affected by the angle alpha and kappa chord,¹² as curvature is centered on the corneal vertex, and elevation is referenced to a best-fit sphere, which is referenced to the corneal vertex or the corneal apex, depending on the platforms, and both references are affected by fixation.¹²

One of the limitations of our study is that the dataset was unbalanced, which may have affected the detection accuracy of the KCS group. Furthermore, we used cross-validation due to the relatively small sample size. Further validation of the algorithm with an independent test dataset is preferable, ideally through a multicenter study. Additionally, pachymetry maps are dependent on tear film distribution, and hence, are sensitive to conditions affect-

A



B

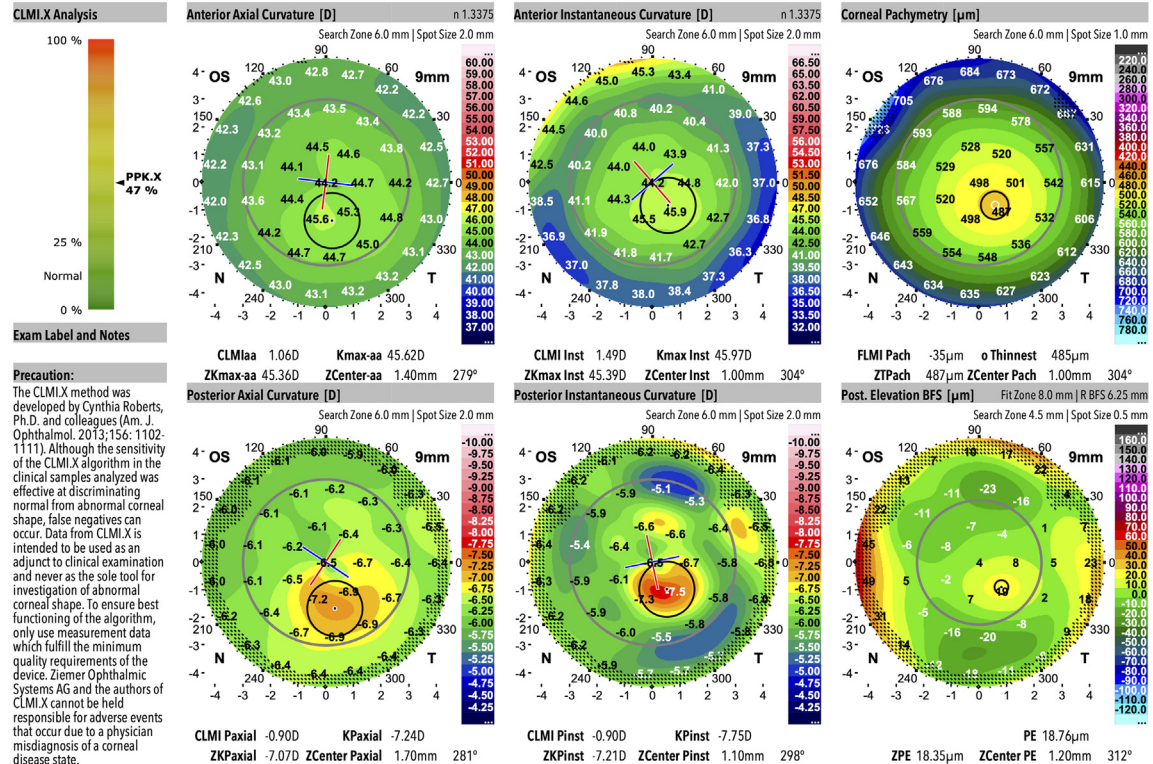


FIGURE 6. The same eye of the same patient in Figure 5, 3 years later. (A) The corneal tomographies reveal slightly increased focal steepening, thinning, and elevation. (B) CLMIX is now abnormal at 47%, indicating an ectatic cornea. (C) The TSPI now labels the eye as “pathologic asymmetry,” indicating keratoconus.

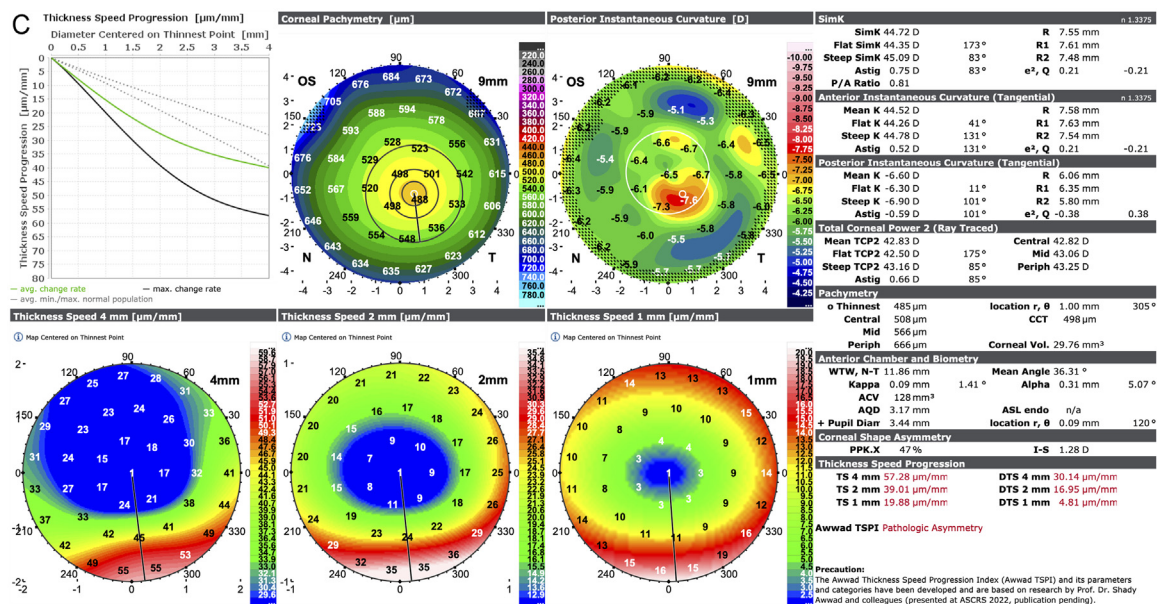


FIGURE 6. Continued

ing the ocular surface, such as dryness, or to suboptimal measurement conditions, such as partial eyelid coverage of the cornea. These factors can impact the quality of measurements thereby affecting the reliability of the TSPI interpretations. It is imperative that future applications of this methodology employ rigorous quality control measures to ensure the acquisition of high-quality measurements.

In future studies, it would be interesting to explore the contribution of the stroma and epithelium to the TSPI by reproducing the CTP map and TSPI on optical coherence tomography with epithelial segmentation. It is also important to note that the TSPI was developed for use on nonoperated corneas and further research is needed to translate its applicability to postlaser vision correction ectasia. The primary aim of this paper was to evaluate the contribution of pachymetry data to the diagnosis of KC using an advanced methodology for pachymetry data analysis. Future endeavors should combine the TSPI with other metrics, such as curvature, elevation, and biomechanical parameters, to further enhance the detection of KCSs.

In conclusion, an enhanced methodology in evaluating asymmetry of corneal pachymetry progression can effectively diagnose KC and improve detection of ectasia susceptibility. The TSPI is an example of such methodology and

combining it with other biomechanical and tomographic metrics would likely increase its robustness.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Shady T. Awwad: Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Bassel Hammoud:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Data curation, Conceptualization. **Jad F. As-saf:** Writing – original draft, Validation, Software, Methodology, Conceptualization. **Lara Asroui:** Writing – original draft, Data curation. **James Bradley Randleman:** Writing – review & editing, Validation, Supervision. **Cynthia J. Roberts:** Writing – review & editing, Supervision, Methodology. **Douglas D. Koch:** Writing – review & editing, Validation, Supervision. **Jawad Kaisania:** Software. **Carl-Joe Mehanna:** Methodology, Data curation. **Shadi Elbassuni:** Supervision.

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